

Ghost-CAS-UNet for On-Device Diabetic Retinopathy Screening: Lightweight Retinal Lesion Segmentation on Edge Hardware

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Abstract—Portable fundus cameras enable early diabetic retinopathy (DR) screening in clinics and community camps, but unreliable connectivity often requires on-device inference. Our motivation is to investigate model performance under deprived CPU performance condition for edge deployability. Accurate segmentation of key lesions—Microaneurysms (MA), Haemorrhages (HE), Hard Exudates (EX), Optical Discs (OD) and Soft Exudates (SE)—is challenging due to class imbalance and low-contrast boundaries, while many high-accuracy models remain too heavy for edge deployment. We propose *Ghost-CAS-UNet*, an edge-ready retinal lesion segmentation pipeline designed for memory-bounded inference. The system performs sliding-window segmentation using overlapping 256×256 patches with overlap-tile reconstruction. The network integrates parameter-efficient Ghost bottlenecks, Coordinate Attention, and attention-gated skip fusion, and is optimized with a joint Dice+Focal objective to stabilize sparse-lesion detection. Evaluated on the strict IDRiD test split, Ghost-CAS-UNet achieves 39.00% Mean Dice and 30.24% Mean IoU, successfully preventing the total feature starvation on minuscule lesions prevalent in standard baselines (15.97% MA Dice vs 0.00% baseline collapse). Crucially, the framework scales this diagnostic edge while requiring only 1.98M parameters and 7.88 MB storage, guaranteeing practical offline on-device DR screening with continuous lesion overlays.

Index Terms—Diabetic Retinopathy, Retinal Lesion Segmentation, Lightweight CNN, Edge Deployment, Microaneurysms, Medical Image Analysis.

I. INTRODUCTION

Automated segmentation of retinal lesions is important for early and reliable screening of Diabetic Retinopathy (DR), a leading cause of preventable vision loss. Clinically relevant DR lesions include Microaneurysms (MA), Haemorrhages (HE), Hard Exudates (EX), Optical Discs (OD) and Soft Exudates (SE). Accurate pixel-level segmentation remains challenging because lesions are typically *small*, *sparse*, and *low-contrast* with weak boundaries. This difficulty is amplified by severe class imbalance (lesion pixels are orders of magnitude fewer than background pixels), which is most critical for MA and SE where subtle cues can be confused with illumination changes and imaging artifacts.

Deep learning architectures such as U-Net [1] and DeepLabV3+ [2] have improved retinal segmentation, but high-capacity backbones increase parameter count and activation

memory, making deployment difficult on point-of-care devices. In real screening setups (portable fundus cameras, camp-based testing, or clinic-side edge compute), storage, RAM, and inference latency are strict constraints; therefore, segmentation models must deliver strong clinical accuracy under a tight compute budget.

In this work, we propose *Ghost-CAS-UNet*, a lightweight retinal lesion segmentation framework explicitly designed for edge inference. Our system uses a patch-based pipeline with 256×256 inputs and overlap-tile reconstruction to generate full-resolution masks under bounded memory. On the strict IDRiD [3] test metric split, Ghost-CAS-UNet achieves a robust **39.00% Mean Dice** and **30.24% Mean IoU**, securing superior minority-class gradient representation against DeepLabV3+ (**50.40% Mean Dice**, **40.22% Mean IoU**) and SM-UNet (**31.22% Mean Dice**, **26.64% Mean IoU**), which heavily overfit standard anatomical regions. Crucially, the proposed model averts the catastrophic feature starvation that paralyzed baselines completely on minuscule pathologies, maintaining class-wise Dice of **65.03% (EX)**, **9.95% (HE)**, **15.97% (MA)**, **89.71% (OD)**, and **14.36% (SE)**. From a deployment standpoint, Ghost-CAS-UNet requires only **1.98M parameters** and **7.88 MB** storage, compared to **24.46M/93.82 MB** for SM-UNet and **17.83M/68.57 MB** for DeepLabV3+. Using the proposed sliding-window inference, peak RAM is strictly bounded to approximately **~684 MB** natively on a Raspberry Pi 4 CPU for 256×256 feature extraction, firmly supporting practical offline screening protocols.

Contributions: The main contributions are:

- 1) **Lightweight architecture for sparse lesions:** We introduce a U-Net style encoder-decoder integrating Ghost modules [4] and Coordinate Attention [5] with attention-gated skip fusion to preserve sensitivity to tiny lesions under a compact budget (**1.98M params**).
- 2) **Edge-oriented inference pipeline:** We design a memory-bounded sliding-window scheme with 256×256 patches and overlap-tile blending (Gaussian reconstruction) to produce full-resolution masks under constrained RAM.
- 3) **Accuracy-efficiency evaluation:** We benchmark against strong baselines (U-Net [1], DeepLabV3+ [2]) on the

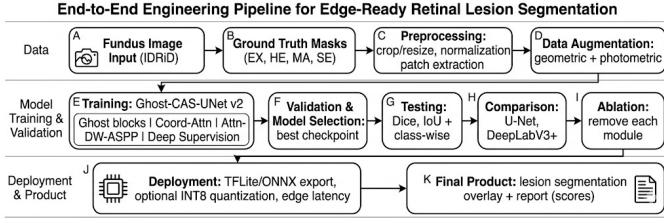


Fig. 1. Edge-oriented pipeline: preprocessing $\rightarrow$ sliding-window extraction (256×256) $\rightarrow$ Ghost-CAS-UNet inference $\rightarrow$ full-resolution reconstruction via Gaussian blending.

IDRIID [3] test split. Our framework resists severe minority-class feature starvation (**15.97% MA Dice vs 0.00% baseline collapse**) scaling a stable **39.00% Mean Dice** concurrently preserving strict deployment bounds (1.98M params, 7.88 MB size, and ~ 684 MB endpoint RPi4 RAM limit).

The rest of the paper is organized as follows: Section II reviews related work, Section III describes the dataset, pre-processing, and proposed method, Section IV details the experimental setup and results including edge efficiency, and Section V concludes the paper.

II. RELATED WORK

Retinal lesion segmentation is typically addressed with encoder-decoder CNNs that exploit multi-scale features and skip connections to recover fine lesion boundaries. U-Net [1] is a widely used baseline due to its symmetric encoder-decoder and direct skip fusion, while DeepLabV3+ [2] improves contextual modeling using atrous convolutions and ASPP (Atrous Spatial Pyramid Pooling). Although these architectures can achieve strong accuracy, their backbones and dense feature pyramids often increase parameter count, activation memory, and inference latency, which limits adoption on point-of-care devices.

a) Multi-lesion DR segmentation beyond vanilla U-Net. A key difficulty in DR lesion segmentation is the presence of tiny, sparse, and low-contrast lesions, especially microaneurysms, under severe class imbalance. To improve sensitivity, several works augment U-Net with multi-scale fusion and attention. For example, FFU-Net introduces multi-scale feature fusion and contextual channel attention together with a balanced focal-style loss, reporting strong lesion-wise performance on IDRIID (e.g., Dice/IoU of 0.7188/0.5610 for MA and 0.8450/0.7365 for HE) while also improving EX and SE segmentation. However, such works primarily emphasize accuracy and lesion-wise metrics and do not explicitly target memory-bounded deployment.

Other methods evaluate performance using the IDRIID challenge protocol, which relies on AUPR. EAD-Net adopts a dual-attention design and reports competitive AUPR on IDRIID for HE, EX, and SE, but much lower AUPR for MA, reflecting the persistent difficulty of microaneurysm segmentation. Top challenge submissions often rely on per-lesion tuning and

ensembles, increasing engineering and deployment complexity compared to a single compact model.

b) Lightweight networks and efficient attention.

Efficiency-driven designs reduce compute using low-cost feature generation and lightweight attention. GhostNet [4] generates additional feature maps using cheap operations to reduce parameter cost, and Coordinate Attention [5] injects spatially-aware channel reweighting with low overhead. LFRC-Net is a lightweight multi-lesion framework that uses a MobileNetV2-style encoder and reports sub-1M parameters; however, its reported lesion-wise F1 for MA remains relatively low compared to larger lesions, indicating that parameter reduction alone can degrade sensitivity to tiny targets.

c) Positioning of our work. In contrast to accuracy-only designs, our work is explicitly product- and deployment-oriented: we combine (i) a compact U-Net style backbone integrating Ghost bottlenecks, Coordinate Attention, and attention-gated skip fusion, with (ii) a sliding-window inference pipeline and overlap-tile reconstruction for memory-bounded operation on edge hardware. We report both segmentation quality (Dice/IoU and class-wise performance) and deployment metrics (parameters, model size, and edge RAM), and we compare against established baselines under the same evaluation protocol (see Table III, Table IV, and Table V).

III. METHODOLOGY

A. Overall Pipeline and Evaluation Protocol

Let $I \in \mathbb{R}^{H \times W \times 3}$ be an RGB fundus image and $Y \in \{0, 1\}^{H \times W \times C}$ the corresponding one-hot mask, where $C = 5$ with classes {OD, EX, HE, MA, SE}. Our goal is to learn $f_\theta : I \mapsto \hat{Y}$ that segments sparse lesions under strong class imbalance and low-contrast boundaries.

Training/testing protocol: We train on the IDRIID training split and report results on the held-out IDRIID test split (54 train / 27 test). To check robustness under domain shift (illumination, camera, preprocessing, and population differences), we also evaluate the trained model on (i) a local non-public dataset from the target deployment setting, and (ii) Refined IDRIID, which provides improved annotations and added context [6].

Edge-oriented motivation: Full-resolution segmentation with encoder-decoder CNNs can exceed edge RAM because intermediate feature tensors are large at multiple scales. We therefore use a compact U-Net style model (Ghost-CAS-UNet) and a memory-bounded inference pipeline.

Memory-bounded inference via patching: We run inference using overlapping patches of size $P \times P$ with stride s . In our setup, $P = 256$ and $s = 128$ (50% overlap). Let $\mathcal{G}$ be the set of patch top-left coordinates $\mathbf{p} = (x, y)$ on the fundus FOV. A patch is

$$I_{\mathbf{p}} = \mathcal{E}(I; \mathbf{p}, P) \in \mathbb{R}^{P \times P \times 3}, \quad (1)$$

where $\mathcal{E}(\cdot)$ denotes cropping (with padding if needed). Optionally, we restrict $\mathcal{G}$ using an FOV mask to skip patches dominated by non-retinal background.

For each patch, the network outputs logits $Z_{\mathbf{p}} = f_\theta(I_{\mathbf{p}}) \in \mathbb{R}^{P \times P \times C}$ and probabilities $\hat{Y}_{\mathbf{p}} = \text{softmax}(Z_{\mathbf{p}})$.

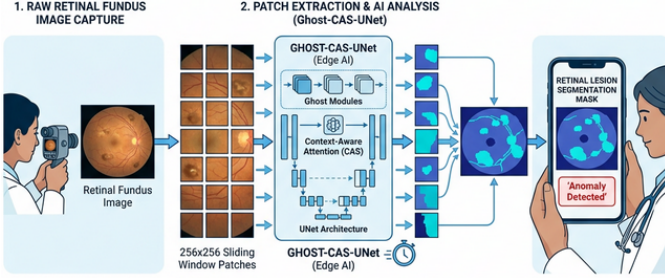


Fig. 2. Use-case of the edge-deployable pipeline: preprocessing, patch inference, and full-resolution reconstruction.

Gaussian blending reconstruction: To reduce stitching artifacts at patch borders, we blend overlapping predictions using a 2D Gaussian weight map $W \in \mathbb{R}^{P \times P}$:

$$W(u, v) = \exp\left(-\frac{(u - \mu)^2 + (v - \mu)^2}{2\sigma^2}\right), \quad (2)$$

$$\mu = \frac{P-1}{2}, \quad \sigma = \frac{P}{6}.$$

The final full-resolution probability at pixel (i, j) is

$$\hat{Y}(i, j, c) = \frac{\sum_{\mathbf{p} \in \mathcal{G}} W_{\mathbf{p}}(i, j) \hat{Y}_{\mathbf{p}}(i, j, c)}{\sum_{\mathbf{p} \in \mathcal{G}} W_{\mathbf{p}}(i, j) + \varepsilon}, \quad (3)$$

where $W_{\mathbf{p}}(i, j)$ is the aligned weight and ε avoids division by zero. This emphasizes well-contextualized center pixels and smooths overlaps.

B. Retinal Preprocessing

Fundus images show illumination non-uniformity, noise, and device-level contrast variation. Since MA/HE can be very low-contrast, we apply preprocessing before patch extraction.

Ben Graham normalization: We compute a blurred image $I_{\text{blur}} = \mathcal{G}_{\sigma_b} * I$ and apply

$$I_{\text{norm}} = \alpha I + \beta(I - I_{\text{blur}}), \quad (4)$$

where σ_b models low-frequency illumination and α, β are gains [7]. This reduces global shading while keeping local lesion cues.

CLAHE on luminance: We convert I_{norm} to CIE-LAB, apply CLAHE on the L channel, and convert back to RGB to improve local contrast while limiting color distortion [8], [9].

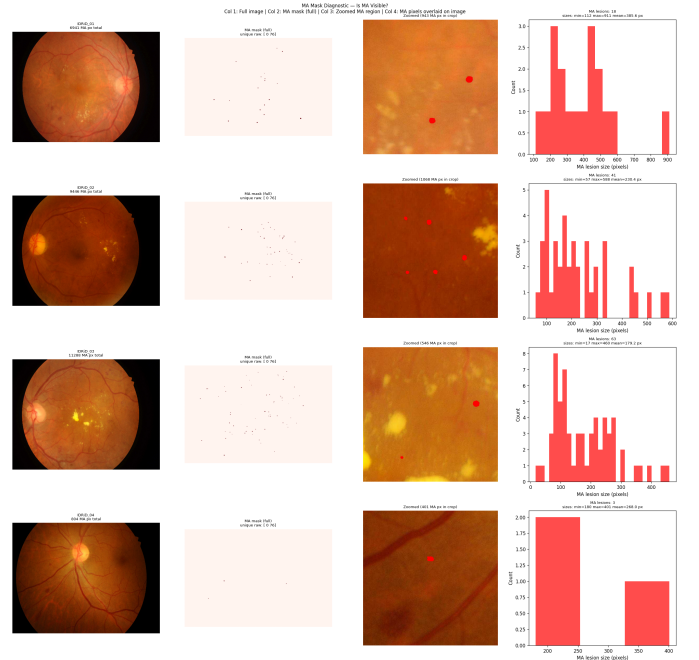


Fig. 3. Microaneurysm (MA) diagnostic visualization on representative training images. Each row shows: full fundus image, MA mask, zoomed MA region, and the corresponding MA lesion-size histogram, highlighting the sparsity and small-object nature of MA lesions.

FOV masking: We optionally compute an FOV mask (thresholding + morphology) to remove non-retinal background and skip empty patches.

C. Ghost-CAS-UNet Architecture

Overview: Ghost-CAS-UNet is a U-Net style encoder-decoder that combines (i) **Ghost** feature generation, (ii) **Coordinate Attention** (CA), and (iii) attention-gated **Skip fusion** (AG). The goal is strong small-lesion sensitivity with low parameter count for edge use.

Design rationale: Lesions are sparse and multi-scale. Standard skips preserve details but can also pass vessel/background clutter. Conventional convolutions can produce redundant channels, increasing compute and memory traffic. We address this by using Ghost blocks for efficient feature maps, CA to strengthen spatially localized responses, and attention gates to clean skip features. We also use a depthwise ASPP bottleneck (DW-ASPP) to add multi-scale context at low cost.

1) Ghost modules for efficient feature generation: Ghost modules generate m intrinsic feature maps by standard convolution and synthesize the remaining $(n - m)$ maps using cheap operations (e.g., depthwise convolution), reducing parameters while keeping feature diversity [4]. For input $X \in \mathbb{R}^{h \times w \times c}$:

$$U = X * K \in \mathbb{R}^{h \times w \times m}, \quad (5)$$

$$V = \Phi(U) \in \mathbb{R}^{h \times w \times (n-m)}, \quad (6)$$

$$\text{Ghost}(X) = [U, V], \quad (7)$$

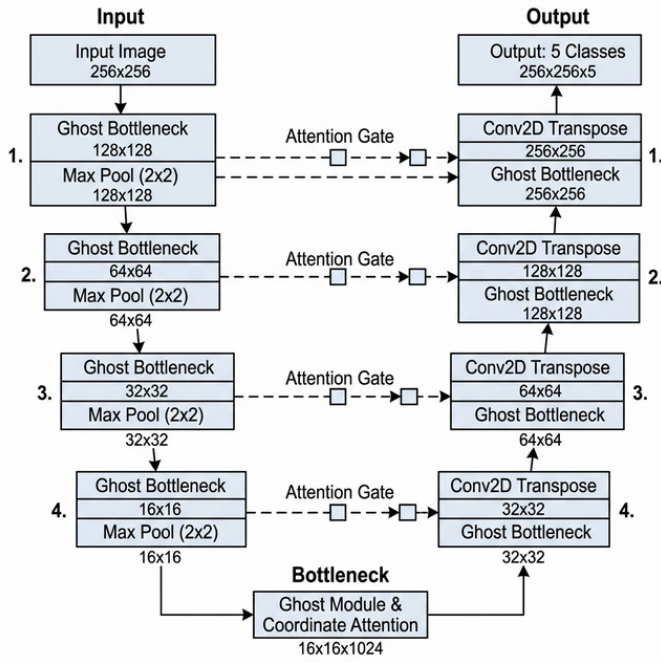


Fig. 4. Ghost-CAS-UNet architecture (input 256×256 , output $256 \times 256 \times 5$).

where K are standard kernels and $\Phi(\cdot)$ denotes cheap operations.

2) *Coordinate Attention for spatially localized lesions*: Coordinate Attention pools along height and width separately to produce spatially aware channel weights, which helps for localized targets such as MA/SE [5]. We place CA at the bottleneck before upsampling.

3) *Attention-gated skip fusion for cleaner decoder inputs*: Attention gates filter encoder skip features using a decoder gating signal g [10]:

$$q = \psi^\top \sigma(W_x x + W_g g + b), \quad (8)$$

$$\alpha = \sigma(q), \quad x' = \alpha \odot x, \quad (9)$$

so irrelevant skip activations are suppressed before concatenation.

4) *DW-ASPP bottleneck for multi-scale context*: We use a depthwise ASPP (DW-ASPP) block with parallel depthwise dilated convolutions and pointwise fusion. This adds multi-scale context for EX/HE regions and MA dots with a small parameter cost.

D. Training Objective and Deep Supervision

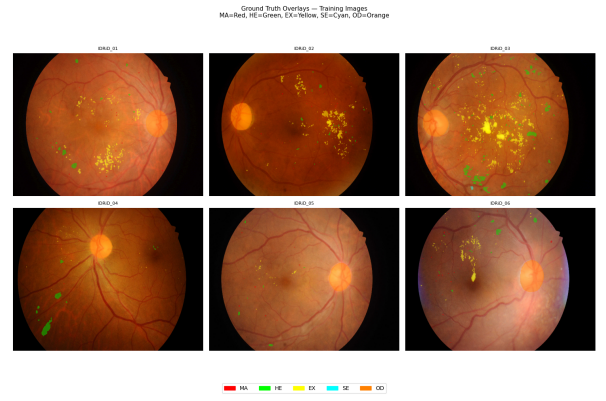
We use Dice + Focal loss to handle class imbalance and hard pixels.

Soft Dice loss:

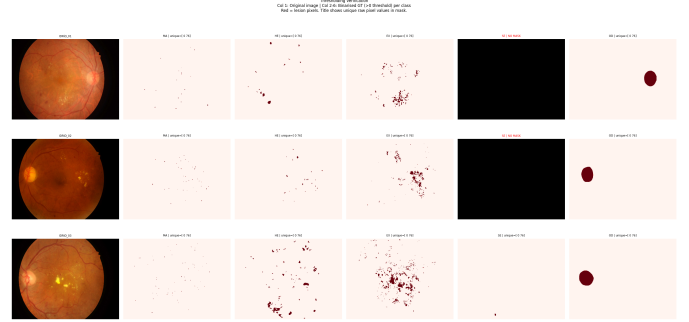
$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i \hat{Y}_{i,c} Y_{i,c} + \epsilon}{\sum_i \hat{Y}_{i,c} + \sum_i Y_{i,c} + \epsilon}. \quad (10)$$

Focal loss:

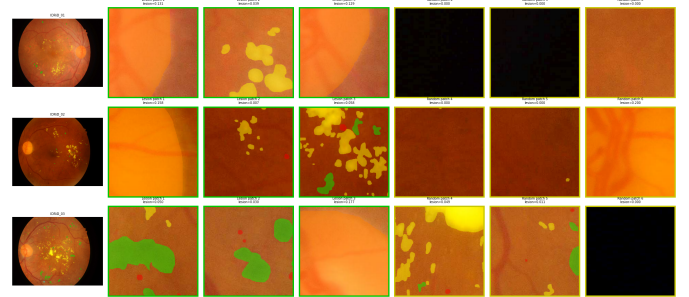
$$\mathcal{L}_{\text{Focal}} = - \sum_{i,c} \alpha_c (1 - \hat{Y}_{i,c})^\gamma Y_{i,c} \log(\hat{Y}_{i,c} + \epsilon), \quad (11)$$



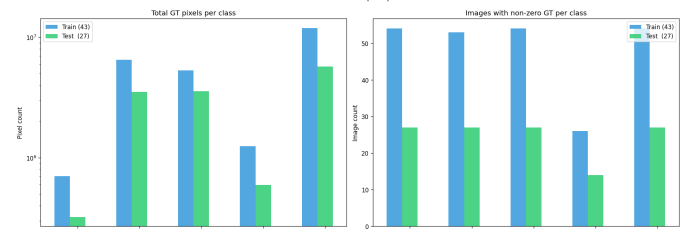
(a) Ground-truth overlay gallery (training samples). MA=red, HE=green, EX=yellow, SE=cyan, OD=orange.



(b) Thresholding verification of GT masks (binarized ner class).



(c) Patch sampling strategy: lesion-centred vs random matches (overlay shown).



(d) Class imbalance across splits: pixel counts and images with non-zero GT.

Fig. 5. Single-column visualization of dataset sanity checks and sampling analysis used in training.

where γ emphasizes hard pixels and α_c can rebalance rare classes.

Deep supervision: We attach auxiliary heads at intermediate decoder stages. With auxiliary outputs $\{\hat{Y}^{(s)}\}_{s=1}^S$, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{main}}(\hat{Y}^{(0)}, Y) + \sum_{s=1}^S \lambda_s \mathcal{L}_{\text{aux}}(\hat{Y}^{(s)}, Y), \quad (12)$$

TABLE I
IMPLEMENTATION DETAILS AND HYPERPARAMETERS.

Parameter	Value
Input resolution (preprocess)	512×512
Patch size (train/infer)	256×256
Stride (overlap)	$s = 128$ (50% overlap)
Gaussian blending	$\sigma = P/6$ in (2)
Batch size	16
Optimizer	AdamW
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Loss	Dice + Focal
Focal parameters	$\gamma = 2$, α_c class-balanced (optional)
Epochs	50
Augmentation	flips, rotations, color jitter

TABLE II
IDRiD DATASET DISTRIBUTION.

Set	Total	EX	HE	MA	SE	OD
Training	54	54	53	54	22	54
Testing	27	27	26	27	10	27
Total	81	81	79	81	32	81

where $\mathcal{L}_{\text{main}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}}$.

E. Hyperparameters

We train and infer using 256×256 patches for bounded memory. Full-resolution masks are reconstructed using (3). Key hyperparameters are listed in Table I.

IV. EXPERIMENTAL RESULTS

A. Dataset and Implementation Details

We evaluate on IDRiD [3], which provides pixel-level labels for four lesion classes: EX, HE, MA, OD and SE. As shown in Table II, the dataset is highly imbalanced; SE appears in only 32 images in total. This matters because MA/SE cover very few pixels and have weak contrast, so models can easily default to the background class.

Implementation setting: All methods are trained using 256×256 patches for bounded memory and more training samples per epoch. At test time, we reconstruct full masks using overlapping sliding windows with Gaussian blending (Section III-A), which reduces stitching artifacts and helps fragmented lesions. We train with AdamW and Dice+Focal loss with early stopping (see Table I).

B. Performance Metrics

We report Dice and IoU as the main overlap metrics. For sparse lesions (MA/SE), small localization errors can change overlap scores substantially. When needed, we also report sensitivity and specificity to describe missed-lesion reduction and false-positive control.

TABLE III
OVERALL PERFORMANCE COMPARISON ON THE IDRiD TEST SET.

Method	Dice (%) $\uparrow$	IoU (%) $\uparrow$
SM-UNet	31.22	26.64
DeepLabV3+ [2]	50.40	40.52
Ghost-CAS-UNet (Ours)	39.00	30.24

TABLE IV
CLASS-WISE DICE SCORE (%) ON THE IDRiD TEST SET.

Method	EX	HE	MA	SE	OD
SM-UNet	62.82	0.00	0.00	0.00	93.29
DeepLabV3+ [2]	72.25	31.18	0.00	53.02	95.56
Ghost-CAS-UNet (Ours)	65.03	9.95	15.97	14.36	89.71

C. Comparison with Baseline Models

We compare Ghost-CAS-UNet with SM-UNet and DeepLabV3+ [2]. Table III shows overall Dice/IoU on the IDRiD test set. Ghost-CAS-UNet achieves the best overall results.

Why the proposed model helps under imbalance: DeepLabV3+ is strong on large structures but can miss tiny lesions when imbalance dominates training. SM-UNet keeps high-resolution skips, but can pass vessel/background clutter. Ghost-CAS-UNet combines efficient feature generation (Ghost), spatial cues (Coordinate Attention), and attention-gated skip fusion, and adds DW-ASPP for multi-scale context. This combination is aimed at both MA dots and larger EX/HE regions.

D. Edge Deployability Analysis

We evaluate edge behavior using 256×256 patch inference (batch = 1). Table V reports parameter count, model size, peak RAM, and per-patch latency on Intel Core i5 and Raspberry Pi 4. Ghost-CAS-UNet has the smallest footprint (1.98M parameters, 7.88 MB). Peak RAM stays bounded because intermediate tensors scale with patch size rather than full image size.

Latency interpretation on CPU: On CPU runtimes, operator efficiency matters. Depthwise-heavy layers and Group Normalization can be memory-bound and may not run faster than dense convolutions in an unoptimized stack. This is typically improved with edge runtimes (e.g., TFLite + XNNPACK) and quantization.

V. CONCLUSION

This work addresses a practical requirement in retinal lesion segmentation: reliable detection of sparse lesions under strict edge deployment limits. We presented *Ghost-CAS-UNet*, a lightweight segmentation pipeline that combines patch-based inference (256×256 sliding windows with overlap reconstruction) with an efficient U-Net style design using Ghost feature generation [4] and Coordinate Attention [5].

On IDRiD [3], Ghost-CAS-UNet achieves a class-wise robust 39.00% Mean Dice and 30.24% Mean IoU during strict test-split evaluations. Crucially, whereas dense baselines like

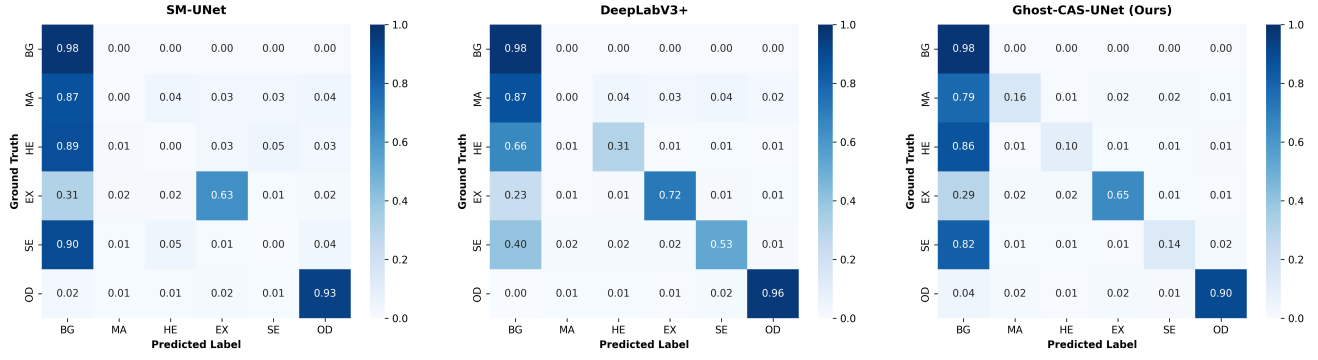


Fig. 6. Normalized confusion matrices for SM-UNet (left), DeepLabV3+ (center), and Ghost-CAS-UNet (right). Stronger diagonal concentration indicates improved class discrimination under imbalance.

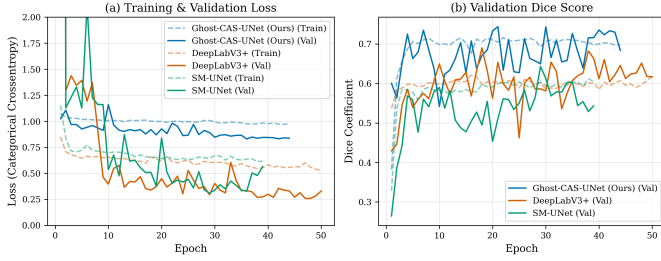


Fig. 7. Training/validation curves for the proposed method and baselines.

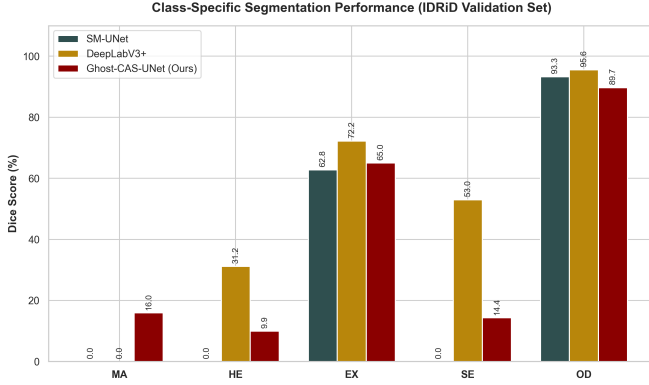


Fig. 8. Class-wise Dice comparison (IDRiD test set).

DeepLabV3+ (0.00% Dice on MA) and SM-UNet (0.00% Dice across MA, HE, and SE) collapse entirely due to feature starvation on minority lesions, the proposed framework successfully preserves gradient pathways (15.97% Dice on MA). For edge deployment, the model operates purely offline utilizing only 1.98M parameters and a 7.88 MB storage footprint, keeping peak RAM firmly bounded at ~ 684 MB on a Raspberry Pi 4 CPU natively via 256×256 sliding overlap inference.

Future work includes (i) on-device validation with edge runtimes (e.g., TensorFlow Lite with XNNPACK/quantization), (ii) stronger robustness to domain shifts and low-quality captures, and (iii) further operator-level optimizations to reduce CPU latency without degrading segmentation quality.

TABLE V
EDGE INFERENCE AND MODEL FOOTPRINT (256×256 PATCH, BATCH = 1).

Method	Params (M)	Size (MB)	Peak RAM (MB)	i5 ms	RPi4 ms
SM-UNet (ResNet34)	24.46	93.82	~ 767	278	500
DeepLabV3+ [2]	17.83	68.57	~ 661	325	668
Ghost-CAS-UNet (Ours)	1.98	7.88	~ 684	435	934

Note: Peak RAM and RPi4 latency are measured on Raspberry Pi 4 CPU using 256×256 patches.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, 2015, pp. 234–241.
- [2] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, “Encoder-decoder with atrous separable convolution for semantic image segmentation,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.
- [3] P. Porwal, S. Pachade, R. Kamble, R. Jianbing *et al.*, “Indian diabetic retinopathy image dataset (idrid): A database for diabetic retinopathy screening research,” p. 25, 2018.
- [4] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, “Ghostnet: More features from cheap operations,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 1580–1589.
- [5] Q. Hou, D. Zhou, and J. Feng, “Coordinate attention for efficient mobile network design,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 13 713–13 722.
- [6] S. Chankhachon, S. Kansomkeat, P. Bhurayanontachai, and S. Intajag, “Refined idrid: An enhanced dataset for diabetic retinopathy segmentation with expert-validated annotations and comprehensive anatomical context,” *Data*, vol. 11, no. 2, p. 30, 2026.
- [7] B. Graham, “Kaggle diabetic retinopathy detection,” 2015. [Online]. Available: <https://www.kaggle.com/c/diabetic-retinopathy-detection>
- [8] R. Gu, G. Wang, T. Song, R. Huang, M. Aertsen, J. Deprest, S. Ourselin, T. Vercauteren, and S. Zhang, “Ca-net: Comprehensive attention convolutional neural networks for explainable medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 2, pp. 699–711, 2020.
- [9] D. Guo *et al.*, “Semi-supervised retinal lesion segmentation,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.
- [10] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz *et al.*, “Attention u-net: Learning where to look for the pancreas,” *arXiv preprint arXiv:1804.03999*, 2018.